Variance Reduction Techniques for Monte Carlo Pricing of European Call Options

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Abstract

Monte Carlo simulation is one of many methods for pricing financial derivatives, such as European call options, but its slow convergence rate makes it nearly impossible to use in practice. This paper compares four Monte Carlo estimators for pricing a European call option under the Black–Scholes model: basic Monte Carlo, antithetic variates, control variates, and Quasi–Monte Carlo using Sobol low-discrepancy sequences. Using repeated simulations across multiple sample sizes, we measure the standard deviation of pricing errors relative to the analytical Black–Scholes value. Antithetic variates provide acceptable improvement, control variates yield substantially lower variance, and Quasi–Monte Carlo delivers the most impressive results, with variance reductions of up to two orders of magnitude. These results demonstrate the effectiveness of advanced sampling techniques in improving Monte Carlo efficiency.

1 Introduction

Monte Carlo simulation is famously used in quantitative finance to price derivatives and assess risk. The method approximates expectations by averaging the simulated outcomes of a stochastic model. Although Monte Carlo methods are very general and can be applied in widely different aspects, such as high-dimensional or path-dependent situations, they suffer from a fundamental limitation: the root-mean-square error decreases only at the rate $O(N^{-1/2})$, where N is the number of simulated paths [4, 3].

Variance reduction techniques aim to increase efficiency by - given in the name - reducing the variance of the Monte Carlo estimator, allowing a more desired accuracy to be reached with fewer simulations [10]. In this work, we study three classical variance reduction methods [4, 10, 9]:

- antithetic variates,
- control variates,
- Quasi–Monte Carlo (QMC) using Sobol low-discrepancy sequences.

We apply these techniques to the pricing of a European call option under the Black–Scholes model. Since a closed-form solution is available, we can quantify the accuracy of each estimator by comparing it with the analytical price. Our goal is not to propose new methods, but instead to compare standard variance reduction techniques in a straightforward and reproducible way.

2 Model and Option Pricing Framework

2.1 Geometric Brownian Motion Under Risk-Neutral Measure

We consider a financial market in which the price of the underlying asset $(S_t)_{t\geq 0}$ follows a simple geometric Brownian motion under the risk-neutral measure.

The dynamic of S_t is given by

$$S_T = S_0 \exp\left((r - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\, Z\right),$$

where S_0 is the initial asset price, r is the continuously compounded risk-free rate, σ is the volatility of the simulation, T is the time to maturity, and $Z \sim N(0, 1)$ is a standard normal random variable [7, 4].

Under the risk-neutral measure, the expected growth rate of the discounted asset price is set to zero, and the drift of S_t is r. This ensures that the discounted asset price process is a martingale and is consistent with no-arbitrage pricing.

2.2 European Call Payoff and Discounted Payoff

We study a European call option with strike K and maturity T. Its payoff at maturity is

$$\max(S_T - K, 0).$$

The discounted payoff under the risk-free rate r is

$$X = e^{-rT} \max(S_T - K, 0).$$

In Monte Carlo simulations, each path generates one realization X_i of this random variable X.

2.3 Black–Scholes Price and Baseline Monte Carlo Estimator

Under the Black–Scholes assumptions, the arbitrage-free price of the European call option is given by the well-known formula [7]

$$C_{\text{BS}} = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2),$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$

and Φ is the cumulative distribution function of the standard normal distribution.

The basic Monte Carlo estimator of the option price is

$$\hat{C}_N = \frac{1}{N}\sum_{i=1}^{N} X_i,$$

where $X_1, \ldots, X_N$ are independent samples of X. This estimator is unbiased:

$$\mathbb{E}[\hat{C}_N] = \mathbb{E}[X].$$

Its variance is

$$\text{Var}(\hat{C}_N) = \frac{\text{Var}(X)}{N},$$

hence, the standard deviation (standard error) of the estimator decreases as $1/\sqrt{N}$ [1, 4].

3 Variance Reduction Techniques

3.1 Antithetic Variates

Antithetic variates profit from the symmetry of the standard normal distribution. Instead of simulating independent samples $Z_1, Z_2, \ldots,$ we simulate pairs $(Z, -Z)$ and compute the respective payoffs of $f(Z)$ and $f(-Z)$, where

$$f(Z) = e^{-rT} \max(S_0 \exp((r - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z) - K, 0).$$

The antithetic estimator for each pair is

$$\hat{C}_{\text{anti}} = \frac{1}{2}\left(f(Z) + f(-Z)\right).$$

Because Z and $-Z$ are negatively correlated in terms of their effect on the payoff, the average $\hat{C}_{\text{anti}}$ ends up being less variable than $f(Z)$ alone. Formally, if we define $X = f(Z)$ and $Y = f(-Z)$, then the variance of the antithetic estimator is

$$\mathrm{Var}\left(\frac{X+Y}{2}\right) = \frac{1}{4}\left(\mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)\right).$$

Since $\mathrm{Var}(X) = \mathrm{Var}(Y)$ and typically $\mathrm{Cov}(X,Y) < 0$, the overall variance decreases [10, 4].

3.2 Control Variates

Control variates use a correlated auxiliary random variable whose expectation is known both analytically and computationally. In this study, we use the terminal stock price S_T as the control variate [10, 1]:

$$Y = S_T, \qquad \mathbb{E}[Y] = S_0 e^{rT}.$$

The main idea is to adjust the payoff X by subtracting a multiple of $(Y - \mathbb{E}[Y])$:

$$C_{\text{cv}} = X - \beta(Y - \mathbb{E}[Y]).$$

The parameter β controls the strength of the correction. The optimal choice (in the sense of minimizing variance) is

$$\beta^* = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)}.$$

Practically, we estimate β using sample covariance and variance computed from the Monte Carlo data.

Intuitively, when Y is higher than its expected value, the option payoff X also tends to be high. Subtracting $\beta(Y - \mathbb{E}[Y])$ removes part of this common tendency, reducing the variability of the adjusted payoff C_{cv} and hence reducing the variance of the estimator.

3.3 Quasi–Monte Carlo Using Sobol Sequences

Quasi–Monte Carlo (QMC) methods replace pseudo-random sampling with low-discrepancy sequences, such as Sobol sequences, that fill the unit interval $[0,1]$ more evenly than random points. Assume $u_1, \ldots, u_N$ be Sobol points in $[0,1]$. We convert them into standard normal variates using the inverse cumulative distribution function [9, 2]:

$$Z_i = \Phi^{-1}(u_i),$$

and then simulate S_T and the discounted payoff X_i as in the previous basic Monte Carlo method.

Because Sobol points avoid clustering and large gaps, they deliver a more uniform coverage of the integration domain than arbitrary sampling. This usually leads to faster convergence than the $O(N^{-1/2})$ rate of standard Monte Carlo. In order to estimate variability across several simulations, we can use scrambled Sobol sequences, which preserve the low-discrepancy mechanism while introducing more randomness between independent runs [3, 2].

4 Experimental Setup

We consider a European call option with the following parameters:

- $S_0 = 100$ (initial stock price),
- $K = 100$ (strike price),
- $r = 0.05$ (risk-free rate),
- $\sigma = 0.2$ (volatility),
- $T = 1$ year (time to maturity).

As per usual, the Black–Scholes formula is used to compute the reference price C_{BS}. For each Monte Carlo method (basic, antithetic, control variates, and Quasi–Monte Carlo), we estimate the option price for a range of sample sizes N, including $N = 2^{10}, 2^{12}, 2^{14}, 2^{16}, 2^{18}$, and 2^{20}. The underlying simulation pipelines and random sampling distributions were structured using the NumPy [5] and SciPy [8] packages.

For each value of N, we repeat the simulation 30 times to estimate the standard deviation of the pricing error

$$\hat{C}_N - C_{\mathrm{BS}}.$$

We then plot the standard deviation of the error versus N on log–log axes using Matplotlib [6] to compare convergence rates and efficiency between methods.

5 Results

Table 1: Convergence Standard Deviation (σ_{error}) Across Scale Dimensions (N)

N (Samples)	Basic MC	Antithetic	Control Variate	Quasi-MC (Sobol)
2^{10}	0.53721	0.34015	0.19133	0.01212
2^{12}	0.23424	0.16547	0.08612	0.00341
2^{14}	0.09689	0.06536	0.04384	0.00070
2^{16}	0.06186	0.04296	0.02614	0.00013
2^{18}	0.02825	0.01919	0.01027	0.00003
2^{20}	0.01307	0.01163	0.00553	0.00001

The findings demonstrate a distinct performance among the computed methodology frameworks. As shown in Table 1,...

5.1 Basic Monte Carlo

The basic Monte Carlo estimator displays the expected behavior: the standard deviation of the error decreases approximately in proportion to $1/\sqrt{N}$. On a log–log plot, this appears as a line with slope close to $-1/2$. However, even for large values of N, the variance remains high. This is the fundamental inefficiency of Monte Carlo for high-accuracy pricing.

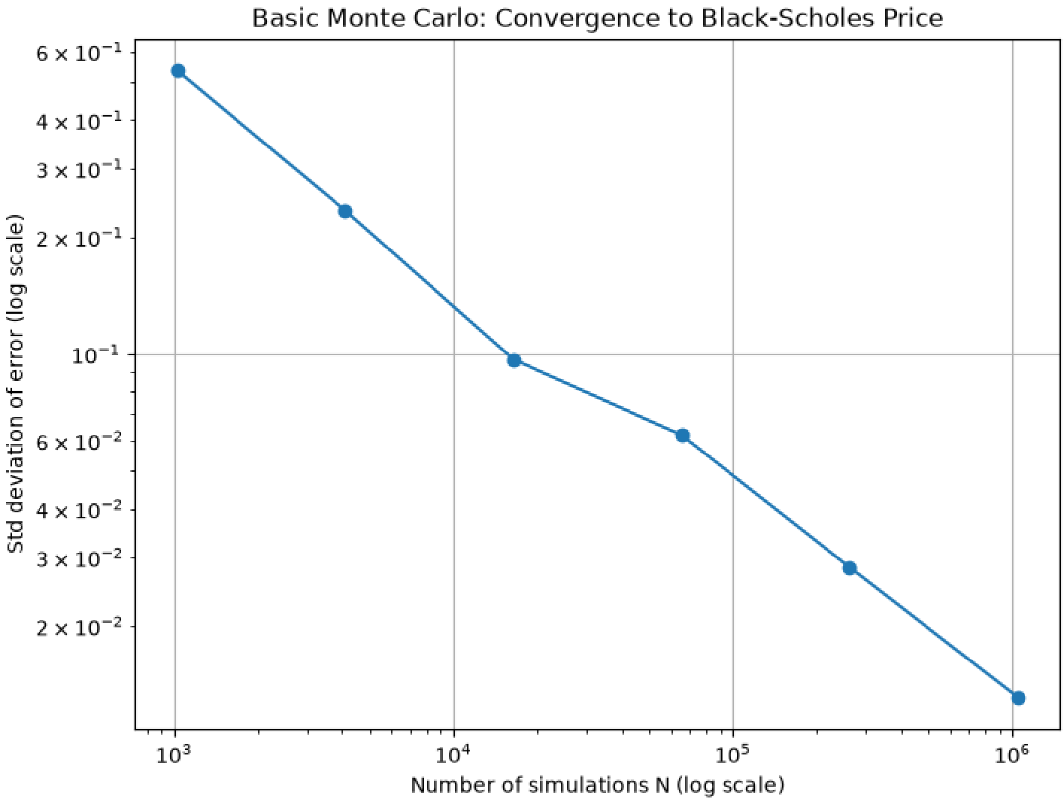

Figure 1: Error convergence of the basic Monte Carlo estimator. The pricing error decreases at the expected rate $N^{-1/2}$, but the variance remains high for large values of N.

5.2 Antithetic Variates

The antithetic variates method reduces the variance by consistently producing lower standard deviations of the error compared to basic Monte Carlo. Across the tested values of N, the variance reduction factor is ordinarily around 1.5 to 2. The convergence rate, with respect to N, remains essentially the same, but the level of the error curve is translated downward.

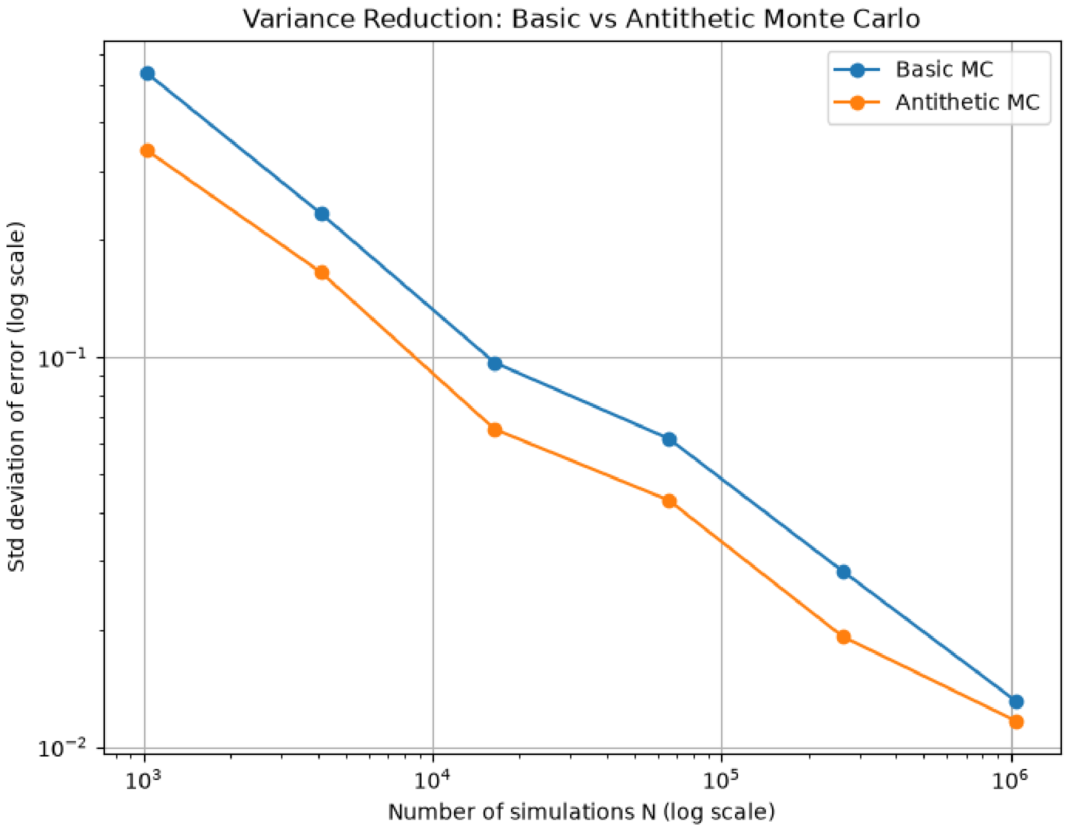

Figure 2: Variance reduction using antithetic variates. Pairing Z with $-Z$ reduces estimator variance by introducing negative correlation, shifting the entire error curve downward while maintaining the same $N^{-1/2}$ convergence rate.

5.3 Control Variates

Control variates deliver a much larger reduction in variance. Using S_T as the control and estimating β from the sample data, the standard deviation of the error is reduced by a factor of about 4 to 6 relative to basic Monte Carlo for the same N. This further confirms that the payoff is strongly correlated with the terminal stock price and that control variates are effective in this application.

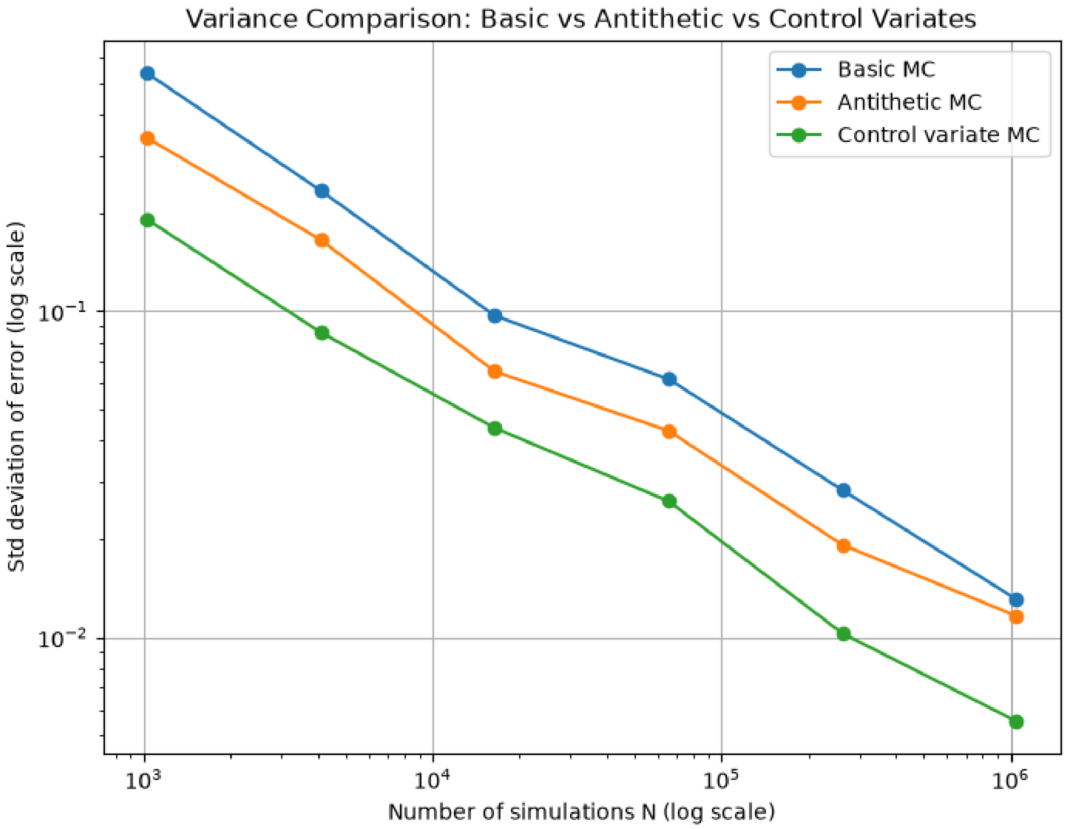

Figure 3: Variance reduction from control variates using the terminal stock price S_T as the control. This method achieves a strong variance reduction factor (typically 4–6×), significantly outperforming both basic and antithetic Monte Carlo.

5.4 Quasi–Monte Carlo

The Quasi–Monte Carlo estimator using Sobol sequences achieves the lowest variance and the most satisfying results among all methods. The standard deviation of the error decreases much more rapidly than for standard Monte Carlo, and for large N the error approaches quasi-deterministic levels. To highlight an example, at $N = 2^{16}$, the QMC estimator is roughly two orders of magnitude more accurate than the basic Monte Carlo estimator, and about an order of magnitude more accurate than the control variate estimator. This demonstrates its high computational efficiency for complex derivative pricing frameworks and applications.

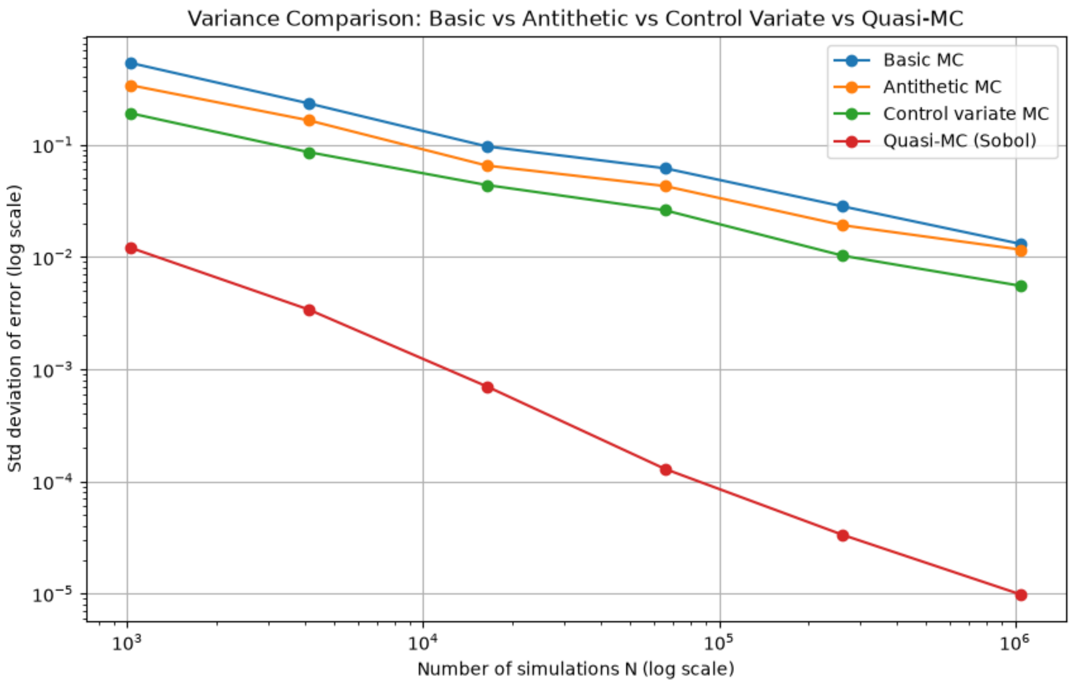

Figure 4: Comparison of all variance reduction methods. Sobol Quasi–Monte Carlo achieves the lowest error overall, demonstrating near-deterministic convergence rates and outperforming basic Monte Carlo by 1–2 orders of magnitude.

6 Discussion

The numerical demonstrations clearly illustrate the benefits of variance reduction techniques in Monte Carlo option pricing. Although basic Monte Carlo is very simple to implement, it is computationally expensive when high accuracy is desired. Antithetic variates offer a modest and easy-to-implement improvement. Control variates, when an effective control variable is available, yield much larger efficiency gains.

Quasi–Monte Carlo methods, using Sobol low-discrepancy sequences, however, provide the most solid performance in this study. Covering the integration domain more uniformly than random sampling, QMC shows dramatic reduction sampling errors and has the strongest potential to achieve the accuracy of millions of standard Monte Carlo samples with practically null function evaluations when compared to the basic Monte Carlo.

When applying mathematics in financial derivatives, the choice of technique depends on the expected payoffs and computational constraints. For simple payoffs with known analytic properties, control variates and QMC are particularly attractive to implement. For more complex or path-dependent derivatives, combinations of these methods are highly recommended and should be studied.

7 Conclusion

This paper compared several variance reduction techniques for the Monte Carlo pricing of a European call option under the Black–Scholes model. Antithetic variates provided consistent but acceptable improvements. Control variates provided a substantial reduction in variance by exploiting the correlation between the payoff and the underlying asset price. Quasi–Monte Carlo using Sobol sequences delivered the highest efficiency, with variance reductions of up to two orders of magnitude compared to basic Monte Carlo.

These results highlight the importance of variance reduction in practical Monte Carlo applications and suggest that Quasi–Monte Carlo methods, in particular, deserve careful and thoughtful consideration in quantitative finance implementations.

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Code Availability

The complete Python experimentation, including the simulation pipelines, variance reduction algorithms, and plotting scripts used to generate the results in this paper, is openly available on GitHub at: `https://github.com/ayhamalashwal/Variance-Reduction-Techniques-for-Monte-Carlo-Pricing-of-European-Call-Options`.